

Predicting local DED track geometry from thermal history

Hierarchical condition baselines, causal multiscale descriptors, physically consistent boundaries, and nested four-track validation

0.163 mm four-track width MAE	13.1% lower than direct Ridge	0.148 mm mean boundary MAE	91.4% conditional coverage
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Executive summary

A laser track is a spatial signal, not one average width. We align each active thermal frame to physical x , extract the local track center and connected left/right profilometry boundaries, and evaluate every track as an untouched condition. The promoted nested selector reduces track-balanced width MAE from 0.187 mm to 0.163 mm (13.1%), worst-track MAE by 28.9%, and boundary MAE by 17.6%.

The key modeling change is to predict condition-level baseline center and log-width from track summaries, then predict local residuals from frame history. This directly addresses cross-power mean shifts and over-smoothed local signals.

Problem formulation and data alignment

For each valid x location, targets are width, center, left boundary, and right boundary. Profilometry cross-sections are robustly detrended; connected 30%-of-peak crossings around the bead define the boundaries. Thermal frames map to x using 10 mm/s at 50 fps (0.2 mm/frame). Frames farther than 0.10 mm from valid geometry are excluded, and only the 24-96 mm steady-state region is scored.

Leakage controls

- Outer leave-one-track-out tests hold out Tracks 8, 10, 14, and 21 in turn.
- Feature family, estimator, preprocessing, and interval calibration are chosen inside the other three tracks by inner leave-one-track-out validation.
- Thermal history is causal: current and earlier frames only. Headline scores are unweighted means of per-track outer metrics.

Generative AI use

OpenAI Codex assisted with code review, tests, debugging, and document layout. All numbers come from tracked analysis code and saved outer-fold predictions; AI did not supply or alter experimental measurements.

Method: preserve condition shifts and local variation

Thermal descriptors. Instantaneous features cover hot-region area and axes, temperature percentiles, thermal mass, gradients, asymmetry, cooling-tail area/decay, centroid velocity, and pool-shape change. Causal 5-, 10-, and 20-frame windows add means, slopes, changes, persistence above fixed temperature thresholds, and history-availability flags. Robust within-track normalization isolates local deviations without exposing held-out geometry.

Constrained multi-output prediction. Track thermal summaries predict baseline center and baseline log-width. Local features jointly predict center and log-width residuals. Exponentiating log-width guarantees positive width; left/right are reconstructed from one shared center and width, so boundaries remain ordered.

Low-capacity model ladder. Inner folds compare Ridge, elastic net, partial least squares, spline-Ridge, and a Gaussian process. Within 0.02 mm of the best inner MAE, selection favors lower variation-scale error and higher residual correlation. Spline-Ridge with normalized compact thermal features wins three outer folds; Track 14 independently chooses normalized multiscale Ridge.

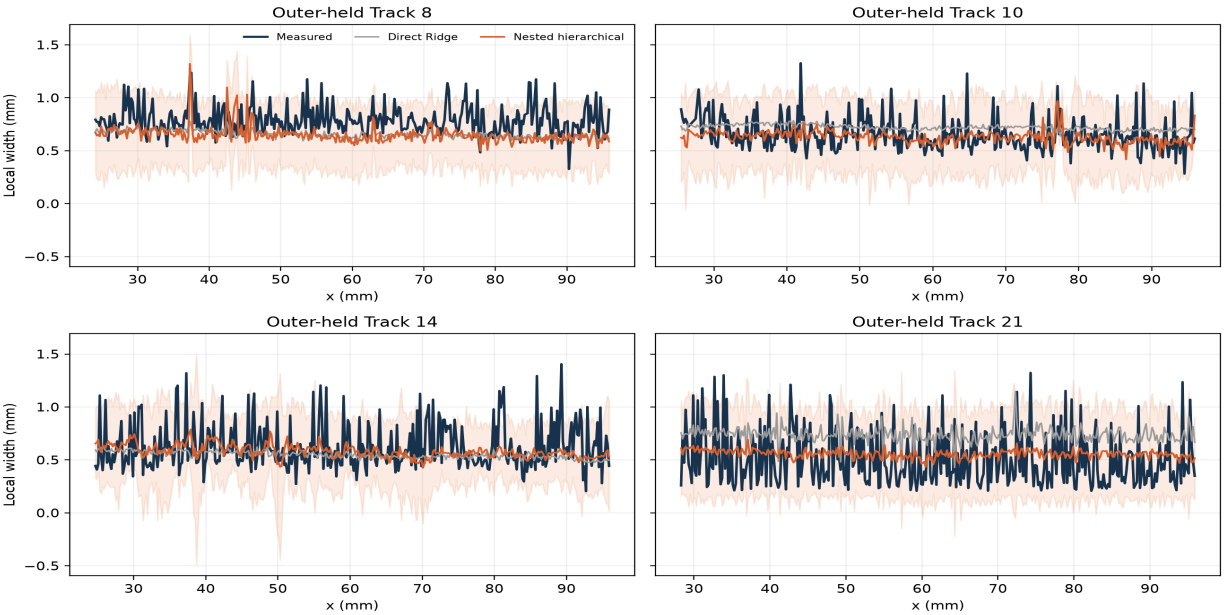


Figure 1. Measured and predicted local width for four untouched outer tracks. Each panel is produced by a model selected without that track's labels.

Metric	Direct Ridge	Hierarchical selector	Change
Track-balanced width MAE	0.187 mm	0.163 mm	-13.1%
Worst-track width MAE	0.308 mm	0.219 mm	-28.9%
Mean left/right boundary MAE	0.180 mm	0.148 mm	-17.6%
Residual correlation	0.055	0.124	2.26x
Predicted/measured variation std.	0.214	0.356	+0.142

Uncertainty, interpretation, and limits

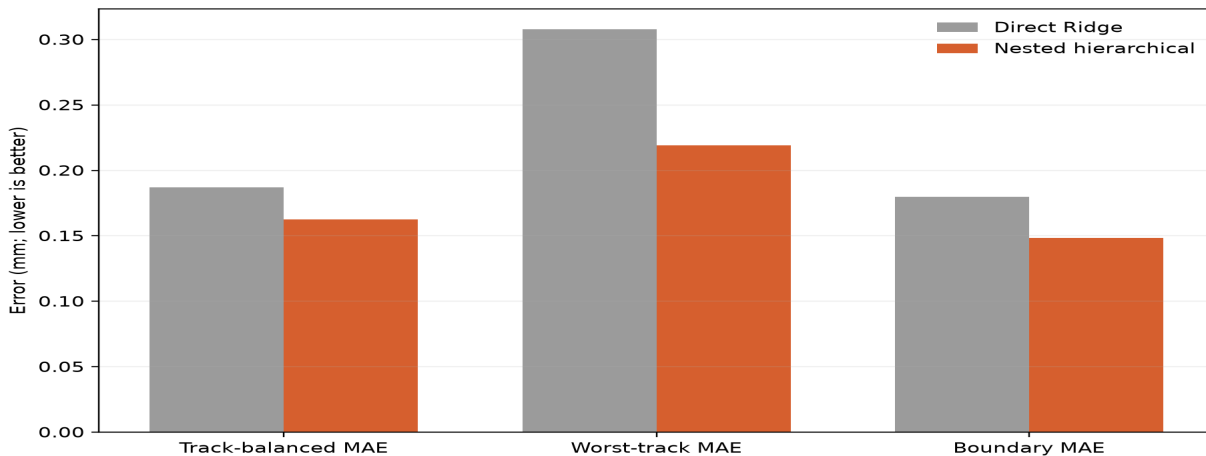


Figure 2. The promoted model improves accuracy, boundary position, and local variation fidelity under the same four-track outer protocol.

Conditional uncertainty

Normalized conformal intervals scale by predicted local difficulty. Conditional coverage is 91.4% with 0.738 mm mean width, versus 94.2% and 0.824 mm for a fixed global interval. Mean width expands from 0.610 mm in easy regions to 0.912 mm in difficult regions.

Interpretable links and substrate evidence

Track-level hot area, maximum temperature, thermal mass, and cooling-tail summaries anchor the condition baseline. Normalized pool shape, temperature, asymmetry, and recent history contribute local residual information. These are predictive associations. The available SEM is post-process; after masking the processed center, SEM is selected in zero outer folds. We therefore make no causal pre-process substrate claim.

Limitations

- Track-balanced R-squared remains -0.34; the result is not ready for closed-loop control.
- Only four independent tracks are available, so model ranking and interval coverage remain uncertain across new plates and recipes.
- The earlier 0.139 mm Track 21 result is a historical tuned split. Under the stronger nested protocol, Track 21 MAE is 0.219 mm.
- Genuine pre-process SEM or surface measurements registered to physical x are needed to distinguish substrate-driven from process-driven variation.

Conclusion

Separating condition-level geometry from local residual variation produces a more accurate, less over-smoothed, physically consistent predictor. The next decisive study should add powers, plates, repeats, and registered pre-process surface measurements, then test the locked pipeline on blind tracks.

Reproduction and sources

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PYTHONPATH=src LOKY_MAX_CPU_COUNT=1 MPLBACKEND=Agg .venv/bin/python
scripts/run_improvement_experiments.py --raw-dir /path/to/data --cache-dir
/path/to/cache --output-dir results/improved_submission
Dataset: doi:10.5281/zenodo.21285367; paper: arXiv:2607.07965; locked metrics:
results/improved_submission/metrics.json.
```